

BENBECULA, United KingdomWMO: **030220**Lat: **57.481N**Lon: **7.363W**Elev: **19**StdP: **14.69**Time Zone: **0.00 (EUW)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	28.4	31.6	23.2	17.5	33.0	26.0	19.9	35.0	45.2	43.5	40.7	43.8	4.9	70	0.803

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	68.1	60.5	64.7	58.3	62.7	57.6	61.5	66.0	59.9	63.6	58.7	61.9	12.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.3	75.5	63.0	58.6	73.4	61.9	57.2	69.9	60.0	27.4	65.5	26.2	63.2	25.4	62.2	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 38.9	33.6	29.9	24.1	74.8	4.1	3.5	21.2	77.3	18.7	79.4	16.4	81.4	13.4	83.9		
(5)			23.8	64.6	3.8	3.0	21.1	66.7	18.9	68.5	16.7	70.2	14.0	72.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.5	41.9	41.2	42.4	46.1	49.7	54.1	57.1	56.8	54.4	50.1	45.3	42.5
	DBStd	6.98	4.03	4.06	3.98	4.16	4.35	3.41	2.88	2.46	3.11	3.83	4.59	5.31
	HDD50	1351	250	248	235	128	55	3	0	0	4	45	148	235
	HDD65	6017	715	668	700	567	473	326	246	254	317	461	592	698
	CDD50	810	0	0	0	10	47	128	219	211	137	49	7	2
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	5	0	0	0	0	2	2	1	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	14.9	17.5	15.7	15.9	14.4	13.6	12.8	12.5	13.1	14.8	15.6	15.9	16.9
	PrecAvg	50.6	6.0	4.3	4.2	3.0	2.5	2.8	3.2	3.9	4.2	5.4	5.4	5.4
(15) Precipitation	PrecMax	66.7	10.3	10.0	8.5	4.8	6.3	5.2	5.6	10.4	8.2	9.2	8.9	9.4
(16) PrecMin		37.0	2.6	1.8	0.8	1.2	0.2	0.9	1.4	1.7	1.3	2.0	3.6	1.2
(17) PrecStd		6.7	1.9	2.0	1.6	0.9	1.4	1.1	1.0	1.8	1.6	2.1	1.2	2.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.6	52.0	55.3	64.0	71.4	71.8	72.6	69.8	64.7	60.7	55.5	54.5
		MCWB	49.6	48.5	49.7	52.6	59.6	62.1	63.1	61.4	58.5	56.8	52.2	52.4
	2%	DB	50.1	50.0	51.5	57.2	64.2	66.4	68.0	64.7	62.2	58.1	53.8	52.2
		MCWB	48.9	48.8	47.9	50.6	55.0	59.3	61.4	59.5	57.9	54.7	51.2	50.2
	5%	DB	48.9	48.5	49.8	53.8	59.7	62.8	64.5	62.6	60.6	57.1	53.1	51.5
		MCWB	47.7	47.4	47.6	48.5	53.2	57.0	59.1	58.2	56.9	54.0	50.7	49.6
	10%	DB	48.1	47.8	48.3	52.0	57.0	60.4	62.4	61.0	59.1	55.6	51.8	50.0
		MCWB	46.9	46.7	46.6	47.8	51.7	55.6	58.2	57.1	55.9	52.6	49.4	48.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.4	49.9	50.3	53.4	60.0	62.8	64.1	62.8	60.2	57.5	53.5	53.2
		MCDB	50.9	50.3	52.2	60.4	67.7	69.6	70.3	66.8	63.0	60.0	54.5	54.6
	2%	WB	49.1	48.7	48.9	51.2	56.9	59.7	62.0	60.3	58.9	55.8	52.4	51.2
		MCDB	49.4	49.2	50.0	55.5	63.1	65.5	66.5	63.5	61.2	58.0	53.7	52.4
	5%	WB	47.9	47.6	47.6	50.1	54.0	57.5	60.1	59.3	57.8	54.7	51.1	49.9
		MCDB	48.5	48.2	48.7	52.8	58.6	62.0	63.7	62.1	59.9	56.7	52.5	50.7
	10%	WB	46.7	45.9	46.6	49.0	52.2	55.9	58.7	58.1	56.7	53.4	49.9	48.4
		MCDB	47.7	46.8	48.0	51.4	56.0	59.4	61.6	60.7	58.8	55.4	51.3	49.4
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	7.8	8.7	10.1	11.3	10.1	9.2	8.7	8.4	8.3	7.8	7.7
		MCDBR	6.7	6.6	9.6	13.2	16.7	15.6	14.3	11.2	10.0	9.3	7.3	8.0
		MCWBR	7.3	6.7	7.7	8.5	9.9	9.7	8.7	7.2	7.3	7.7	6.8	8.9
	5% WB	MCDBR	6.6	5.9	7.7	10.2	14.7	14.0	12.2	9.5	8.7	8.0	7.0	7.6
		MCWBR	7.5	6.7	7.2	7.5	9.6	9.3	8.1	6.9	7.3	7.8	7.3	8.7
(40) Clear-Sky Solar Irradiance	taub		0.354	0.361	0.376	0.386	0.388	0.376	0.379	0.370	0.372	0.354	0.338	0.330
	taud		2.124	2.232	2.283	2.313	2.366	2.445	2.449	2.479	2.430	2.355	2.199	2.108
	Ebn at Noon		146	209	243	261	269	274	271	266	245	212	153	118
	Edh at Noon		18	25	31	35	35	33	32	29	27	22	17	14
(44) All-Sky Solar Radiation	RadAvg		116	306	674	1174	1554	1582	1447	1150	753	390	155	78
	RadStd		13	30	51	82	116	157	100	75	35	38	14	9

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page